



Convergence of Discrete Exterior Calculus Approximations for Poisson Problems

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Abstract

Discrete exterior calculus (DEC) is a framework for constructing discrete versions of exterior differential calculus objects, and is widely used in computer graphics, computational topology, and discretizations of the Hodge–Laplace operator and other related partial differential equations. However, a rigorous convergence analysis of DEC has always been lacking; as far as we are aware, the only convergence proof of DEC so far appeared is for the scalar Poisson problem in two dimensions, and it is based on reinterpreting the discretization as a finite element method. Moreover, even in two dimensions, there have been some puzzling numerical experiments reported in the literature, apparently suggesting that there is convergence without consistency. In this paper, we develop a general independent framework for analyzing issues such as convergence of DEC without relying on theories of other discretization methods, and demonstrate its usefulness by establishing convergence results for DEC beyond the Poisson problem in two dimensions. Namely, we prove that DEC solutions to the scalar Poisson problem in arbitrary dimensions converge pointwise to the exact solution at least linearly with respect to the mesh size. We illustrate the findings by various numerical experiments, which show that the convergence is in fact of second order when the solution is sufficiently regular. The problems of explaining the second order convergence, and of proving convergence for general p -forms remain open.

Keywords Discrete exterior calculus (DEC) · Discrete Hodge operator · Structure preserving · Convergence

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1 Introduction

The main objective of this paper is to establish convergence of discrete exterior calculus approximations on unstructured triangulations for the scalar Poisson problem in general dimensions. There are several approaches to extending the exterior calculus to discrete spaces; what we mean by *discrete exterior calculus* (DEC) in this paper is the approach put forward by Hirani et al., cf. [5, 10]. Since its conception, DEC has found many applications and has been extended in various directions, including general relativity [7], electrodynamics [16], linear elasticity [26], computational modeling [6], port-Hamiltonian systems [15], digital geometry processing [4], Darcy flow [12], and the Navier–Stokes equations [13]. However, a rigorous convergence analysis of DEC has always been lacking; as far as we are aware, the only convergence proof of DEC so far appeared is for the scalar Poisson problem in two dimensions, and it is based on reinterpreting the discretization as a finite element method, cf. [8]. In the current paper, we develop a general independent framework for analyzing issues such as convergence of DEC without relying on theories established for other discretization methods, and demonstrate its usefulness by proving that DEC solutions to the scalar Poisson problem in arbitrary dimensions converge to the exact solution as the mesh size tends to 0.

Developing an original framework to study the convergence of DEC allows us to explore to what extent the theory is compatible in the sense of [1]. At the turn of the millennium, compatibility appeared as a conducive paradigm for stability. In this spirit, we here reproduce at the discrete level a standard variational technique in the analysis of PDEs based on the use of the Poincaré inequality.

In what follows, we would like to describe our results in more detail. Let K_h be a family of n -dimensional completely well-centered simplicial complexes triangulating an n -dimensional polytope $\mathcal{P}$ in $\mathbb{R}^n$, with the parameter $h > 0$ representing the diameter of a typical simplex in the complex. Let $\Delta_h = \delta_h d_h + d_h \delta_h$ be their associated discrete Hodge–Laplace operators, where the discrete exterior derivatives d_h and the codifferentials $\delta_h = (-1)^{n(k-1)+1} \star_h d_h \star_h$ are defined as in [5, 10] up to a sign. Denote by $C^k(K_h)$ the space of k -cochains over K_h . In this paper, we study the convergence of solutions $\omega_h \in C^k(K_h)$ solving discrete Hodge–Laplace Dirichlet problems of the form

$$\Delta_h \omega_h = R_h f \tag{1}$$

in K_h , where f and g are differential forms, and R_h is the de Rham operator, cf. [23]. It is shown in [24, 25] that under *very* special symmetry assumptions on K_h , the consistency estimate

$$\|\Delta_h R_h \omega - R_h \Delta \omega\|_\infty = O(h^2)$$

holds for sufficiently smooth functions ω . However, as the numerical experiments from [14, 24] revealed, this does *not* hold for general triangulations. In Sect. 3, we show that a common shape regularity assumption on K_h can only guarantee

$$\|\Delta_h R_h \omega - R_h \Delta \omega\| = O(1) + O(h) \quad (2)$$

for sufficiently smooth functions ω , in both the maximum $\|\cdot\|_\infty$ and discrete L^2 -norm $\|\cdot\|_h$.

Although the consistency estimate (2) is not adequate for the Lax–Richtmyer framework, by making use of a special structure of the error term itself, which basically amounts to measuring consistency in a discrete H^{-1} -norm, we are able to establish convergence for 0-forms in general dimensions. Namely, we prove in Sect. 4 that the approximations $u_h \in C^0(K_h)$ obtained from solving (1) still satisfy

$$\|\omega_h - R_h \omega\|_h = O(h) \quad \text{and} \quad \|d(\omega_h - R_h \omega)\|_h = O(h), \quad (3)$$

where $\omega \in C^2(\mathcal{P})$ is the solution of the corresponding continuous Poisson problem with source term f and boundary condition g . We remark that a convergence proof in two dimensions can be obtained by reinterpreting the discrete problem as arising from affine finite elements, cf. [8,22], which shows that the first quantity in (3) should indeed be $O(h^2)$ if the exact solution is sufficiently regular. This is consistent with the numerical experiments from [14], and moreover, in Sect. 5, we report some numerical experiments in three dimensions, which suggest that one has $O(h^2)$ convergence in general dimensions. Therefore, our $O(h)$ convergence result for 0-forms should be considered only as a first step towards a complete theoretical understanding of the convergence behavior of discrete exterior calculus. Apart from explaining the $O(h^2)$ convergence for 0-forms, proving convergence for general p -forms remains as an important open problem.

This paper is organized as follows. In the next section, we review the basic notions of discrete exterior calculus, not only to fix notations, but also to discuss issues such as the boundary of a dual cell in detail to clarify some inconsistencies existing in the current literature. Then in Sect. 3, we treat the consistency question, and in Sect. 4, we establish stability of DEC for the scalar Poisson problem. Our main result (3) is proved in Sect. 4. We end the paper by reporting on some numerical experiments, in Sect. 5.

2 Discrete Environment

We review the basic definitions involved in discrete exterior calculus that are needed for the purposes of this work. Readers unacquainted with these discrete structures are encouraged to go over the material covered in the important works cited below. The shape regularity condition that we impose on our triangulations is discussed in Sect. 2.1, and a new definition for the boundary of a dual cell is introduced in Sect. 2.3.

2.1 Simplicial Complexes and Regular Triangulations

The basic geometric objects upon which DEC is designed are borrowed from algebraic topology. While the use of cube complexes is discussed in [2], here we consider

simplices to be the main building blocks of the theory. The following definitions are given in [5] and can also be found in any introductory textbook on simplicial homology.

By a k -simplex in $\mathbb{R}^n$, we mean the k -dimensional convex span

$$\sigma = [v_0, \dots, v_k] = \left\{ \sum_{i=0}^k a_i v_i \mid v_i \in \mathbb{R}^n, a_i \geq 0, \sum_{i=0}^k a_i = 1 \right\},$$

where $v_0, \dots, v_k$ are affinely independent. We denote its circumcenter by $c(\sigma)$. Any ℓ -simplex τ defined from a nonempty subset of these vertices is said to be a face of σ , and we denote this partial ordering by $\sigma \succ \tau$. We write $|\sigma|$ for the k -dimensional volume of a k -simplex σ and adopt the convention that the volume of a vertex is 1. The plane of τ is defined as the smallest affine subspace of $\mathbb{R}^n$ containing τ and is denoted by $P(\tau)$. A simplicial n -complex K in $\mathbb{R}^n$ is a collection of n -simplices in $\mathbb{R}^n$ such that:

- Every face of a simplex in K is in K .
- The intersection of any two simplices of K is either empty or a face of both.

It is well-centered if the circumcenter of a simplex of any dimension is contained in the interior of that simplex. If the set of simplices of a complex L is a subset of the simplices of another complex K , then L is called a *subcomplex* of K . We denote by $\Delta_k(L)$ the set of all elementary k -simplices of L . The *star* of a k -simplex $\sigma \in K$ is defined as the set $\text{St}(\sigma) = \{\rho \in K \mid \sigma \prec \rho\}$. $\text{St}(\sigma)$ is not closed under taking faces in general. It is thus useful to define the closed star $\widehat{\text{St}}(\sigma)$ to be the smallest subcomplex of K containing $\text{St}(\sigma)$. We denote the free abelian group generated by a basis consisting of the oriented k -simplices by $C_k(K)$. This is the space of finite formal sums of k -simplices with coefficients in $\mathbb{Z}$. Elements of $C_k(K)$ are called *k -chains*. More generally, we will write $\bigoplus_k C_k(K)$ for the space of finite formal sums of elements in $\bigcup_k C_k(K)$ with integer coefficients. The *boundary* of a k -chain is obtained using the linear operator $\partial: \bigoplus_k C_k(K) \rightarrow \bigoplus_k C_{k-1}(K)$ defined on a simplex $[v_0, \dots, v_k] \in K$ by

$$\partial[v_0, \dots, v_k] = \sum_{i=0}^k (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_k].$$

Any simplicial n -complex K in $\mathbb{R}^n$ such that $\bigcup_{\sigma \in \Delta_n(K)} \sigma = \mathcal{P}$ is called a *triangulation* of $\mathcal{P}$. We consider in this work families of well-centered triangulations K_h in which each complex is indexed by the size $h > 0$ of its longest edge. We write γ_τ for the radius of the largest $\dim(\tau)$ -ball contained in τ . The following condition imposed on K_h is common in the finite element literature, see, e.g., [3]. We suppose there exists a shape regularity constant $C_{\text{reg}} > 0$, independent of h , such that for all simplex σ in K_h ,

$$\text{diam}(\sigma) \leq C_{\text{reg}} \gamma_\sigma.$$

A family of triangulations satisfying this condition is said to be *regular*. Important consequences of this regularity assumption will later follow from the next lemma.

Lemma 2.1 *Let $\{K_h\}$ be a regular family of simplicial complexes triangulating an n -dimensional polytope $\mathcal{P}$ in $\mathbb{R}^n$. Then there exists a positive constant $C_\#$, independent of h , such that*

$$\#\left\{\sigma \in \Delta_n(K_h) : \sigma \in \overline{\text{St}}(\tau)\right\} \leq C_\# \text{ for all } \tau \in \Delta_k(K_h).$$

Proof If $\tau \prec \sigma$, then σ contains all the vertices of τ . It is thus sufficient to consider the case where τ is a vertex. Let v be a vertex in K_h , and suppose $\sigma \in \overline{\text{St}}(v)$. Let B_σ be the n -dimensional ball of radius γ_σ contained in σ . The set $A = \bigcup_{\eta \in \Delta_{n-1}(\sigma)} \eta \cap B_\sigma \cap \text{St}(\sigma)$ contains exactly $n + 1$ points in $\mathbb{R}^n$. The argument

$$\theta = \min_A \arctan \left(\frac{\gamma_\sigma}{|x - v|} \right)$$

is thus well-defined, and it satisfies $\theta \geq \arctan(1/C_{\text{reg}})$ by the regularity assumption. Consider an n -dimensional cone of height γ_σ , apex v and aperture 2θ contained in σ . Its generatrix determines a spherical cap V_σ on the hypersphere S_σ of radius h centered at v . The intersection of spherical caps determined by cones contained in distinct simplices is countable. Therefore, there can be at most

$$\frac{\text{vol}(S_\sigma)}{V(\arctan(1/C_{\text{reg}}))} = 2\pi \left(\int_0^{\arctan(1/C_{\text{reg}})} \sin^n(t) dt \right)^{-1}$$

of them, and only so many distinct n -simplices containing v as a result. $\square$

2.2 Combinatorial Orientation and Duality

A triangulation stands as a geometric discretization of a polytopic domain, but it ought to be equipped with a meaningful notion of orientation if a compatible discrete calculus is to be defined on its simplices. We outline below the exposition found in [10].

We felt the need to review the definitions of Sect. 2.1 partly to be able to stress the fact that the expression of a k -simplex σ comes naturally with an ordering of its vertices. Defining two orderings to be equivalent if they differ by an even permutation yields two equivalence classes called *orientations*. The vertices themselves are dimensionless. As such, they are given no orientation. By interpreting a permutation ρ of the vertices in σ as an ordering for the basis vectors $v_{\rho(1)} - v_{\rho(0)}, v_{\rho(2)} - v_{\rho(1)}, \dots, v_{\rho(k)} - v_{\rho(k-1)}$, we see that these equivalence classes coincide with the ones obtained when the affine space $P(\sigma)$ is endowed with an orientation in the usual sense. A simplex is thus oriented by its plane and vice versa. The planes of the $(k - 1)$ -faces of σ inherit an orientation as subspaces of $P(\sigma)$. Correspondingly, we establish that the *induced orientation* by $[v_0, \dots, v_k]$ on the $(k - 1)$ -face $v_0, \dots, \hat{v}_i, \dots, v_k$ is the same as the orientation of $[v_0, \dots, \hat{v}_i, \dots, v_k]$ if i is even, while the opposite orientation is assigned otherwise.

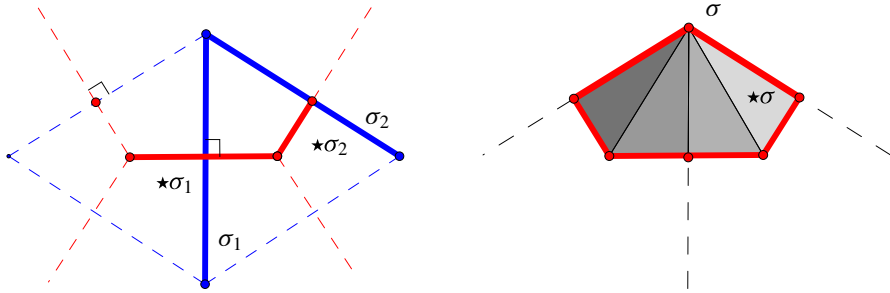


Fig. 1 Boundary and interior faces are perpendicular to their duals. On the left is shown a 1-dimensional complex in blue and its dual in red. On the right, each shade of grey indicates a 2-dimensional simplex in D_h . The dual of the vertex σ is a complex of these simplices, and its boundary is colored in red

Two k -simplices σ and τ in $\mathbb{R}^n$ are hence comparable if $P(\sigma) = P(\tau)$, or if they share a $(k - 1)$ -face. In the first case, the *relative orientation* $\text{sign}(\sigma, \tau) = \pm 1$ of the two simplices is $+1$ when their bases yield the same orientation of the plane and -1 otherwise. The induced orientation on the common face is similarly used to establish relative orientation in the second case. The mechanics of orientation are conveniently captured by the structure of exterior algebra. For example, $v_1 - v_0 \wedge \cdots \wedge v_k - v_{k-1}$ could be used to represent the orientation of $[v_0, \dots, v_k]$. From now on, we will always assume that K_h is well-centered and that all its n -simplices are positively oriented with respect to each other. The orientations of lower dimensional simplices are chosen independently.

We are now ready to introduce the final important objects pertaining to the discrete domain before we move on to the functional aspects of DEC. Recalling that we write $c(\tau)$ for the circumcenter of a simplex τ , we denote by D_h the smallest simplicial n -complex in $\mathbb{R}^n$ containing every simplex of the form $[c(v), \dots, c(\tau), \dots, c(\sigma)]$, where the simplices $v < \cdots < \tau < \cdots < \sigma$ belong to K_h . Let $*$: $\bigoplus_k C_k(K_h) \rightarrow \bigoplus_k C_{n-k}(D_h)$ be the homomorphism acting on $\tau \in K_h$ by

$$*\tau = \sum_{\tau < \cdots < \sigma} \pm_{\sigma, \dots, \tau} [c(\tau), \dots, c(\sigma)],$$

where $\pm_{\sigma, \dots, \tau} = \text{sign}([v, \dots, c(\tau)], \tau) \text{sign}([v, \dots, c(\sigma)], \sigma)$. We define the *oriented circumcentric dual* of K_h by $*K_h = \{*\tau \mid \tau \in K_h\}$ (see Fig. 1). Note that the space $C_{n-k}(*K_h)$ of finite formal sums of $(n - k)$ -dimensional elements of $*K_h$ with integer coefficients is a subgroup of $C_{n-k}(D_h)$ onto which the restriction of $*$ to $C_k(K_h)$ is an isomorphism. In fact, since a simplex whose vertices are $c(\tau), \dots, c(\sigma)$ is oriented in the above formula to satisfy $P(\tau) \times P(c(\tau), \dots, c(\sigma)) \sim \sigma$, we effortlessly extend the definition of $*$ to $\bigoplus_k C_k(*K_h)$ as well by defining $**\tau = (-1)^{k(n-k)} \tau$ for every simplex $\tau \in \Delta_k(K_h)$, $k = 1, \dots, n$.

2.3 Discrete Operators

A strength of discrete exterior calculus is foreseen in the following objects. The theory is built on a natural and intuitive notion of discrete differential forms having a straightforward implementation. We call k -cochains the elements of the space $C^k(K_h) = \text{Hom}(C_k(K_h), \mathbb{R})$, and define $\bigoplus_k C^k(K_h)$, $C^k(*K_h)$ and $\bigoplus_k C^k(*K_h)$ as expected. The *discrete Hodge star* is an isomorphism $\star_h: \bigoplus_k C^k(K_h) \rightarrow \bigoplus_k C^{n-k}(*K_h)$ satisfying

$$\langle \star_h \omega_h, * \sigma \rangle = \frac{|\star \sigma|}{|\sigma|} \langle \omega_h, \sigma \rangle \quad \text{for all } \omega \in K_h. \quad (4)$$

In complete formal analogy, we further impose that

$$\langle \star_h \star_h \omega_h, * * \sigma \rangle = \left(\frac{|\star \sigma|}{|\sigma|} \right)^{-1} \langle \star_h \omega_h, * \sigma \rangle \quad \text{for all } \omega \in K_h.$$

In other words, $\langle \star_h \star_h \omega_h, \sigma \rangle = (-1)^{k(n-k)} \langle \star_h \star_h \omega_h, * * \sigma \rangle = (-1)^{k(n-k)} \langle \omega_h, \sigma \rangle$ for all k -cochains ω_h , and as a consequence $\star_h \star_h = (-1)^{k(n-k)}$ on $C^k(K_h)$. The spaces $C^k(K_h)$ are finite dimensional Hilbert spaces when equipped with the discrete L^2 inner product

$$(\alpha_h, \beta_h)_h = \sum_{\tau \in \Delta_k(K_h)} \langle \alpha_h, \tau \rangle \langle \star_h \beta_h, * \tau \rangle, \quad \alpha_h, \beta_h \in C^k(K_h).$$

We denote the discrete L^2 -norm induced on $C^k(K_h)$ by $\|\cdot\|_h$. A compatible definition of the discrete L^2 -norm on the dual triangulation is also chosen by enforcing the Hodge star to be an isometry. The norm in the following definition is obtained from the inner product:

$$(\star \alpha_h, \star \beta_h)_h = \sum_{\tau \in \Delta_k(K_h)} \langle \star \alpha_h, * \tau \rangle \langle \star_h \star \beta_h, * * \tau \rangle, \quad \star \alpha_h, \star \beta_h \in C^k(*K_h),$$

which is immediately seen to satisfy $(\alpha_h, \beta_h)_h = (\star \alpha_h, \star \beta_h)_h$.

Definition 2.2 The *discrete L^2 -norm* on $C^k(*K_h)$ is defined by

$$\|\star \omega_h\|_h^2 = \sum_{\tau \in \Delta_k(K_h)} \left(\frac{|\star \tau|}{|\tau|} \right)^{-1} \langle \star_h \omega_h, * \tau \rangle^2, \quad \omega_h \in C^k(*K_h).$$

It will also be convenient to measure the largest value taken by a k -cochain as a function of the k -simplices of the triangulation. In that respect, we define the maximum norm of $\omega_h \in C^k(K_h)$ by

$$\|\omega_h\|_\infty := \max_{\tau \in C_k(K_h)} |\langle \omega_h, \tau \rangle|,$$

and in accordance with definition (4),

$$\|\star_h \omega_h\|_\infty = \max_{\tau \in C_k(K_h)} |\langle \star_h \omega_h, \star_h \tau \rangle|.$$

The last step towards a full discretization of the Hodge–Laplace operator is to discretize the exterior derivative. While requiring that

$$\langle d\omega_h, \tau \rangle = \langle \omega_h, \partial\tau \rangle, \quad \text{for all } \tau \in K_h,$$

readily defines a *discrete exterior derivative* $d_h: \bigoplus_k C^k(K_h) \rightarrow \bigoplus_k C^{k+1}(K_h)$ that satisfies Stokes’ theorem, we need to make precise what is meant by the boundary of a dual element if we hope to define the exterior derivative on $C^k(\star K_h)$ similarly. The current literature offers a definition for the boundary of a dual cell whose orientation is compatible with the classical theory of manifolds with boundary only up to a sign. In the following investigation, we fix this sign discrepancy by leveraging the calculus of exterior algebra.

We examine the case of a k -simplex $\tau = \pm_\tau[v_0, \dots, v_k]$ in K_h . It is sufficient to restrict our attention to an n -dimensional simplex σ to which τ is a face. We assume that the orientation of σ is consistent with $(v_1 - v_0) \wedge (v_2 - v_1) \wedge \dots \wedge (v_n - v_{n-1})$, and thereon begin our study. The orientation of τ is represented by $\pm_\tau(v_1 - v_0) \wedge \dots \wedge (v_k - v_{k-1})$. We have seen that enforcing the orientation of its circumcentric dual $\star\tau = \pm_{\star\tau}[c(\tau), c([v_0, \dots, v_{k+1}]), \dots, c(\sigma)]$ to satisfy the definition given in Sect. 2.2 is equivalent to requiring that $\pm_\tau \pm_{\star\tau}(v_1 - v_0) \wedge \dots \wedge (v_n - v_{n-1}) \sim \sigma$. Consequently, $\pm_\tau = \pm_{\star\tau}$ by hypothesis (i.e. the signs agree), and we obtain $\star\tau \sim \pm_\tau(v_{k+1} - v_k) \wedge \dots \wedge (v_n - v_{n-1})$. Since similar equivalences hold for any $(k+1)$ -face $\eta = \pm_\eta[v_0, \dots, v_{k+1}]$ of σ , the orientation induced by η on the face $v_0, \dots, v_k$ is defined to be the same as the orientation of

$$(-1)^{k+1} \pm_\eta \pm_\tau \tau \sim (-1)^{k+1} \pm_\eta (v_1 - v_0) \wedge \dots \wedge (v_k - v_{k-1}).$$

Therefore, choosing $\pm_{\text{old}} = (-1)^{k+1} \pm_\eta \pm_\tau$ makes the induced orientation of $\pm_{\text{old}}\eta$ on that face consistent with the orientation of τ , and yields on the one hand that the orientation of $c(\eta), \dots, c(\sigma)$ as a piece of $\star \pm_{\text{old}} \eta$ is given by

$$(-1)^{k+1} \pm_\tau (\pm_\eta)^2 (v_{k+2} - v_{k+1}) \wedge \dots \wedge (v_n - v_{n-1}).$$

On the other hand, $P(\pm_{\text{new}}(\star \pm_{\text{old}} \eta))$ is a subspace of codimension 1 in $P(\star\tau)$. As such, it inherits a boundary orientation in the usual sense through the assignment

$$P(\star\tau) \sim \pm_{\text{new}}(-1)^{k+1} \pm_\tau \vec{v} \wedge (v_{k+2} - v_{k+1}) \wedge \dots \wedge (v_n - v_{n-1}),$$

where $\vec{v}$ is any vector pointing away from $P(\star\eta)$. Since by hypothesis $\vec{v} = (v_{k+1} - v_k)$ is a valid choice of outward vector, we conclude that the above compatible relation holds if and only if $\pm_{\text{new}} = (-1)^{k+1}$. The claim that the following new definition for the boundary of a dual element is suited for integration by parts rests on the later observation (Fig. 2).

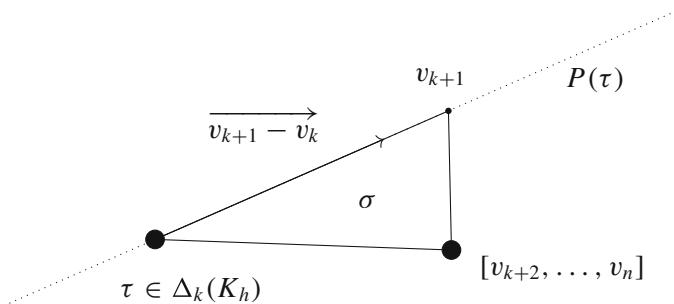


Fig. 2 This figure illustrates the exposition of Sect. 2.3. The plane in which the triangle lies is a representation for the n -dimensional plane of σ , while the dotted line represents $P(\tau)$

Definition 2.3 The linear operator $\partial: \bigoplus_k C_{n-k}(*K_h) \longrightarrow \bigoplus C_{n-k-1}(*K_h)$, which we call the *dual boundary operator*, is defined as the linear operator acting on the dual of $\tau = [v_0, \dots, v_k] \in K_h$ by

$$\partial * \tau = (-1)^{k+1} \sum_{\eta > \tau} * \eta,$$

where η is a $(k+1)$ -simplex in K_h oriented so that the induced orientation of η on the face $v_0, \dots, v_k$ is consistent with the orientation of τ .

Example 2.4 Let $\sigma = [v_0, v_1, v_2]$ be an oriented 2-simplex. The 2-dimensional dual of $\tau = v_0$ is given by

$$* \tau = \pm_{v_0, [v_0, v_1], \sigma} [v_0, c([v_0, v_1]), c(\sigma)] + \pm_{v_0, [v_0, v_2], \sigma} [v_0, c([v_0, v_2]), c(\sigma)],$$

where by definition

$$\begin{aligned} \pm_{v_0, [v_0, v_1], \sigma} &= \text{sign}([v_0, c([v_0, v_1]), c(\sigma)], \sigma) = 1; \\ \pm_{v_0, [v_0, v_2], \sigma} &= \text{sign}([v_0, c([v_0, v_2]), c(\sigma)], \sigma) = -1. \end{aligned}$$

The orientation of the boundary edges of $* \tau$ with endpoints $c([v_0, v_2]), c(\sigma)$ and $c([v_0, v_1]), c(\sigma)$ are compatible with integration by parts if they are assigned an orientation equivalent to the one given in the above expression. Yet, from the definition of the boundary of a dual cell as found in the literature,

$$\begin{aligned} \partial * \tau &= *((-1)[v_0, v_1]) + *((-1)[v_0, v_2]) \\ &= *[v_1, v_0] + *[v_2, v_0] \\ &= \pm_{[v_1, v_0], \sigma} [c(v_0, v_2), c(\sigma)] + \pm_{[v_2, v_0], \sigma} [c(v_0, v_1), c(\sigma)], \end{aligned}$$

where

$$\begin{aligned} \pm_{[v_1, v_0], \sigma} &= \text{sign}([v_0, c([v_1, v_0]), [v_1, v_0]]) \cdot \text{sign}([v_0, c([v_0, v_1]), c(\sigma)], \sigma) \\ &= (-1)(+1) = -1; \end{aligned}$$

$$\begin{aligned}\pm_{[v_2, v_0] \sigma} &= \text{sign}([v_0, c([v_2, v_0])], [v_2, v_0]) \cdot \text{sign}([v_0, c([v_0, v_2]), c(\sigma)], \sigma) \\ &= (-1)(-1) = +1.\end{aligned}$$

Using Definition 2.3, we can now resolutely extend the exterior derivative to

by defining it as the homomorphism satisfying

It is tempting to define the discrete codifferential as the adjoint of the discrete exterior derivative under a suitable inner product. In mimetic finite differences and in some interpretation of finite elements, this can be accomplished by introducing a reconstruction operator (e.g. the Whitney map). However, the discrete codifferential of DEC is built piece by piece using the objects that were designed compatible with the continuous theory.

$$\delta_h = (-1)^{n(k-1)+1} \star_h d_h \star_h.$$
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desirable geometric properties of integration (Stokes' theorem) at the discrete level, but that DEC is also compatible with the dual structure of smooth de Rham complexes. The importance of this fact lies in that it allows to reproduce at the discrete level common analysis techniques based on the variational formulations of Hodge–Laplace problems. The next proposition rely on the new Definition 2.3 for the boundary of the dual cell.

Proposition 2.6 *If $\omega_h \in C^k(K_h)$ and $\eta_h \in C^{k+1}(K_h)$, then*

$$(\mathrm{d}_h \omega_h, \eta_h)_h = (\omega_h, \delta_h \eta_h)_h. \quad (5)$$

Proof It is sufficient to consider the case where $\omega_h(\tau) = 1$ on a k -simplex τ in K_h and vanishes otherwise. On the one hand,

$$(\mathrm{d}_h \omega_h, \eta_h)_h = \sum_{\sigma} \langle \omega_h, \partial \sigma \rangle \langle \star_h \eta_h, \star \sigma \rangle = \langle \omega_h, \tau \rangle \sum_{\sigma \succ \tau} \langle \star \eta_h, \star \sigma \rangle, \quad (6)$$

where σ is a $(k+1)$ -simplex oriented so that it is consistent with the induced orientation on τ . On the other hand, since $\delta_h = (-1)^k \star_h^{-1} \mathrm{d}_h \star_h$ on k -cochains, it follows from Definition 2.3 that

$$(\omega_h, \delta_h \eta_h)_h = (-1)^{k+1} \langle \omega_h, \tau \rangle \langle \mathrm{d}_h \star_h \eta_h, \star \tau \rangle = \langle \omega_h, \tau \rangle \sum_{\sigma \succ \tau} \langle \star \eta_h, \star \sigma \rangle,$$

where σ is oriented as in (6). $\square$

Incidentally, Definition 2.3 also yields a proof of the standard algorithm supplied for the practical calculation of the boundary of a dual cell (see Sect. 5).

3 Consistency

In this section, we develop a framework to address consistency questions for operators such as the Hodge–Laplace operator

$$\Delta_h = \delta_h \mathrm{d}_h + \mathrm{d}_h \delta_h.$$

We begin by proving a preliminary lemma which reduces the question of consistency of the Hodge–Laplace operator to the one of consistency of the Hodge star. We then proceed with a demonstration of the latter, and derive bounds on the norms of various discrete operators.

We denote by $C^r \wedge^k(\mathcal{O})$ the space of differential k -forms ω on an open set $\mathcal{O} \subset \mathbb{R}^n$ such that the function

$$x \mapsto \omega_x(X_1(x), \dots, X_k(x))$$

is r times continuously differentiable for every k -tuples $X_1, \dots, X_k$ of smooth vector fields on $\mathcal{O}$. Given an n -polytope $\mathcal{Q}$, say an n -dimensional subcomplex of K_h , we write

$C^r \Lambda^k(\mathcal{Q})$ for the set of differential k -forms obtained by restricting to $\mathcal{Q}$ a differential form in $C^r \Lambda^k(\mathcal{O})$, where $\mathcal{O}$ is any open set in $\mathbb{R}^n$ containing $\mathcal{Q}$. The ‘exact’ discrete representatives for solutions of the continuous problems corresponding to (1) are constructed using the *de Rham map* R_h defined on $C \Lambda^k(\mathcal{P})$ by

$$\langle R_h \omega, s \rangle = \int_s \omega,$$

where s is a k -dimensional simplex in K_h or the dual of an $(n - k)$ -dimensional one. The next lemma depends on the extremely convenient property that $d_h R_h = R_h d$.

Lemma 3.1 *Given $\omega \in C^2 \Lambda^k(\mathcal{P})$, we have*

$$\begin{aligned} \Delta_h R_h \omega - R_h \Delta \omega &= \star_h d_h (\star_h R_h - R_h \star) d\omega + (\star_h R_h - R_h \star) d\star d\omega \\ &\quad + d_h (\star_h R_h - R_h \star) d\star \omega + d_h \star_h d_h (\star_h R_h - R_h \star) \omega. \end{aligned}$$

Proof Consider the explicit expression

$$\begin{aligned} \Delta_h R_h \omega - R_h \Delta \omega &= (\delta_h d_h + d_h \delta_h) R_h \omega - R_h (\delta d + d\delta) \omega \\ &= (\star_h d_h \star_h d_h R_h - R_h \star d\star) \omega + (d_h \star_h d_h \star_h R_h - R_h d\star d\star) \omega \end{aligned}$$

of the discrete Hodge–Laplace operator. Substituting

$$\begin{aligned} \star_h d_h R_h \omega &= \star_h R_h d\omega = \star_h R_h d\omega - R_h \star d\omega + R_h \star d\omega \\ &= (\star_h R_h - R_h \star) d\omega + R_h \star d\omega \end{aligned}$$

in the first summand, we obtain

$$\begin{aligned} \star_h d_h \star_h d_h R_h \omega - R_h \star d\star d\omega &= \star_h d_h (\star_h R_h - R_h \star) d\omega \\ &\quad + \star_h d_h R_h \star d\omega - R_h \star d\star d\omega \\ &= \star_h d_h (\star_h R_h - R_h \star) d\omega + (\star_h R_h - R_h \star) d\star d\omega. \end{aligned}$$

The second summand can be rewritten as

$$\begin{aligned} (d_h \star_h d_h \star_h R_h - R_h d\star d\star) \omega &= d_h (\star_h d_h \star_h R_h - R_h \star d\star) \omega \\ &= d_h (\star_h d_h \star_h R_h - \star_h R_h d\star + \star_h R_h d\star - R_h \star d\star) \omega \\ &= d_h (\star_h R_h - R_h \star) d\star \omega \\ &\quad + d_h (\star_h d_h \star_h R_h - \star_h R_h d\star) \omega. \end{aligned}$$

The desired equality then follows from the identity

$$(\star_h d_h \star_h R_h - \star_h R_h d\star) \omega = (\star_h d_h \star_h R_h - \star_h d_h R_h \star) \omega = \star_h d_h (\star_h R_h - R_h \star) \omega,$$

which completes the proof. $\square$

Given an n -simplex σ and a k -dimensional face $\tau \prec \sigma$, the subspaces $P(\tau)$ and $P(*\tau)$ of $\mathbb{R}^n$ are perpendicular and satisfy $P(\tau) \times P(*\tau) = P(\sigma)$. Since the discrete Hodge star maps k -cochains on K_h to $(n - k)$ -cochains on the dual mesh $*K_h$, it captures the geometric gist of its classical Hodge operator $\star$, which acts on wedge products of orthonormal covectors as $\star(d\lambda_{\rho(1)} \wedge d\lambda_{\rho(2)} \wedge \cdots \wedge d\lambda_{\rho(k)}) = \text{sign}(\rho) d\lambda_{\rho(k+1)} \wedge d\lambda_{\rho(k+2)} \wedge \cdots \wedge d\lambda_{\rho(n)}$.

We hope to exploit this correspondence to estimate the consistency of $\star_h$. By definition, the de Rham map directly relates the integral of a differential k -form ω over a k -dimensional simplex τ of K_h with the value of its discrete representative at τ . But one can also integrate any differential $(n - k)$ -form, in particular $\star\omega$, over $*\tau$. This suggests that simple approximation arguments could be used to compare $\langle R_h \star\omega, *\tau \rangle$ with $\langle \star_h \omega_h, *\tau \rangle$, since the latter is defined as a scaling of the aforesaid integral $\omega_h = R_h \omega$ by the area factor $|*\tau|/|\tau|$, which is also naturally related to those very integrals under consideration.

The technique we use is best illustrated in low dimensions. Let $p = (p_1, p_2)$ be a vertex of a triangulation K_h embedded in $\mathbb{R}^2$. For a function f on $\mathbb{R}^2$, we have $\star f = f dx \wedge dy$. On the one hand, $\langle R_h f, p \rangle$ is simply the evaluation of f at p . On the other hand, if f is differentiable, then a Taylor expansion around p yields

$$\begin{aligned} \langle R_h \star f, *p \rangle &= \iint_{*p} f dA \\ &= \iint_{*p} f(p) + ((x, y) - (p_1, p_2))^T Df(p) + O(h^2) dA \\ &= |*p|f(p) + O(h^3). \end{aligned}$$

Relying on our convention that $|p| = 1$, we conclude that

$$\langle R_h \star f, *p \rangle - \langle \star_h R_h f, *p \rangle = \langle R_h \star f, *p \rangle - |*p|f(p) = O(h^3). \quad (7)$$

This argument is used in the next theorem as a prototype which we generalize in order to derive estimates for differential forms of higher order in arbitrary dimensions.

Theorem 3.2 *Let σ be an n -simplex, and suppose $\tau \prec \sigma$ is k -dimensional. Then*

$$\langle \star_h R_h \omega, *\tau \rangle = \langle R_h \star \omega, *\tau \rangle + O(h^{n+1}/(\gamma_\tau)^k), \quad \omega \in C^1 \Lambda^k(\sigma).$$

Proof Denote by $\Sigma(k, n)$ the set of strictly increasing maps $\{1, \dots, k\} \longrightarrow \{1, \dots, n\}$. Since $P(\tau) \perp P(*\tau)$ and $P(\tau) \times P(*\tau) = \mathbb{R}^n$, we can choose orthonormal vectors $t_i = x_i - c(\tau)$, $x_i \in \mathbb{R}^n$, such that $\{t_i\}_{i=1, \dots, k}$ and $\{t_i\}_{i=k+1, \dots, n}$ are bases for $P(\tau)$ and $P(*\tau)$ respectively. Let $d\lambda_1, \dots, d\lambda_n$ be a basis for $\text{Alt}^1 \mathbb{R}^n$ dual to $\{t_i\}_{i=1, \dots, n}$. We write vol_τ for $d\lambda_1 \wedge \cdots \wedge d\lambda_k$ and $\text{vol}_{*\tau}$ for $d\lambda_{k+1} \wedge \cdots \wedge d\lambda_n$.

Following the argument which led to (7), we approximate the integral $\langle R_h \omega, \tau \rangle$ of a differential k -form $\omega = \sum_{\rho \in \Sigma(k, n)} f_\rho(d\lambda)_\rho$ over τ using Taylor expansion. This yields

$$\langle R_h \omega, \tau \rangle = \int_{\tau} \sum_{\rho \in \Sigma(k, n)} f_{\rho}(\mathrm{d}\lambda)_{\rho} = \int_{\tau} f_{1, \dots, k} \mathrm{vol}_{\tau} = |\tau| f_{1, \dots, k}(c(\tau)) + O(h^{k+1}).$$

We find similarly that

$$\begin{aligned} \langle R_h \star \omega, \star \tau \rangle &= \int_{\star \tau} \sum_{\rho \in \Sigma(k, n)} f_{\rho} \star (\mathrm{d}\lambda)_{\rho} \\ &= \int_{\star \tau} f_{1, \dots, k} \mathrm{vol}_{\star \tau} = |\star \tau| f_{1, \dots, k}(c(\tau)) + O(h^{n-k+1}). \end{aligned}$$

Combining these two equations, we get

$$\begin{aligned} \langle R_h \star \omega, \star \tau \rangle &= \frac{|\star \tau|}{|\tau|} \langle R_h \omega, \tau \rangle + O(h^{n+1}/(\gamma_{\tau})^k) + O(h^{n-k+1}) \\ &= \langle \star_h R_h \omega, \star \tau \rangle + O(h^{n+1}/(\gamma_{\tau})^k), \end{aligned}$$

which completes the proof. $\square$

Corollary 3.3 *Integrating on the dual mesh, we obtain*

$$\langle \star_h R_h(\star \omega), \tau \rangle = \langle R_h \star(\star \omega), \tau \rangle + O(h^{k+1}), \quad \omega \in C^1 \Lambda^k(\mathcal{P}),$$

when K_h is regular.

Proof Using Theorem 3.2, we find that

$$\begin{aligned} \langle \star_h R_h(\star \omega), \tau \rangle &= (-1)^{k(n-k)} \langle \star_h R_h(\star \omega), \star \star \tau \rangle \\ &= (-1)^{k(n-k)} \left(\frac{|\star \tau|}{|\tau|} \right)^{-1} \langle R_h(\star \omega), \star \tau \rangle \\ &= (-1)^{k(n-k)} \left(\frac{|\star \tau|}{|\tau|} \right)^{-1} \left(\frac{|\star \tau|}{|\tau|} \langle R_h \omega, \tau \rangle + O(h^{n-k+1}) \right) \\ &= \langle R_h \star(\star \omega), \tau \rangle + O(h^{k+1}). \end{aligned} \quad \square$$

Corollary 3.4 *The estimate*

$$\|\star_h R_h \omega - R_h \star \omega\|_{\infty} = O(h^{n-k+1}), \quad \omega \in C^1 \Lambda^k(\mathcal{P}),$$

holds when K_h is regular.

Corollary 3.5 *The estimate*

$$\|\star_h R_h \omega - R_h \star \omega\|_h = O(h), \quad \omega \in C^1 \Lambda^k(\mathcal{P}),$$

holds when K_h is regular.

Proof Using Theorem 3.2, we estimate directly

$$\begin{aligned}\|\star_h R_h \omega - R_h \star \omega\|_h^2 &= \sum_{\tau \in \Delta_k(K_h)} \left(\frac{\star \tau}{\tau}\right)^{-1} \langle \star_h R_h \omega - R_h \star \omega, \star \tau \rangle^2 \\ &\leq C \sum_{\tau \in \Delta_k(K_h)} \left(\frac{\star \tau}{\tau}\right)^{-1} \left(h^{n-k+1}\right)^2 \\ &\lesssim \sum_{\tau \in \Delta_k(K_h)} h^{n+2} \\ &\lesssim h^2,\end{aligned}$$

where the last inequality is obtained from the fact that $\#\{\tau \mid \tau \in \Delta_k(K_h)\} \sim h^{-n}$, which is a consequence of the regularity assumption. $\square$

Finally, the following corollary is of special importance. Its proof is similar to the one used for Corollary 3.5.

Corollary 3.6 *The estimate*

$$\|(\star_h R_h - R_h \star)(\star \omega)\|_h = O(h), \quad \omega \in C^1 \Lambda^k(\mathcal{P}),$$

holds when K_h is regular.

Proof

$$\begin{aligned}\|(\star_h R_h \omega - R_h \star)(\star \omega)\|_h^2 &= \sum_{\tau \in \Delta_k(K_h)} \langle (\star_h R_h \omega - R_h \star)(\star \omega), \tau \rangle \langle \star_h (\star_h R_h \omega - R_h \star)(\star \omega), \star \tau \rangle \\ &= \sum_{\tau \in \Delta_k(K_h)} \frac{|\star \tau|}{|\tau|} \langle (\star_h R_h \omega - R_h \star)(\star \omega), \tau \rangle^2 \\ &\lesssim \sum_{\tau \in \Delta_k(K_h)} h^{n-2k} (h^{k+1}) \\ &\lesssim h^2.\end{aligned}$$

$\square$

As claimed, we see that estimates on the norm of the operators defined in Sect. 2.3 would finally allow one to use the above corollaries to find related bounds on the consistency expression given in Lemma 3.1. In particular, since

$$\Delta_h \omega_h = \delta_h d_h \omega_h, \quad \omega_h \in C^0(K_h), \quad (8)$$

it is sufficient for the Hodge–Laplace problem on 0-forms to evaluate the operator norm of the discrete exterior derivative and of the discrete Hodge star over $C^k(\star K_h)$,

$k = 1, \dots, n$. Namely, from (8) and the proof of Lemma 3.1, the consistency error of the Hodge–Laplace restricted to 0-forms ω can be written as

$$\Delta_h R_h \omega - R_h \Delta \omega = \star_h d_h (\star_h R_h - R_h \star) d\omega + (\star_h R_h - R_h \star) d\star d\omega.$$

The following estimates mainly rely on Lemma 2.1, and on the regularity assumption imposed on K_h . Indeed, for $\omega_h \in \Delta_k(K_h)$, the inequality

$$\|d_h \star_h \omega_h\|_\infty \leq C \|\star_h \omega_h\|_\infty, \quad (9)$$

is a direct consequence of the former, whereas

$$\|\star_h \star_h \omega_h\|_\infty \sim h^{2k-n} \|\star_h \omega_h\|_\infty \quad (10)$$

is immediately obtained from the latter. While

$$\|\star_h\|_{\mathcal{L}((C^k(*K_h), \|\cdot\|_h), (C^{n-k}(K_h), \|\cdot\|_h))} = 1 \quad (11)$$

is merely a statement about the Hodge star being an isometry, the last required estimate is derived from

$$\begin{aligned} \|d_h \star_h \omega_h\|_h^2 &= \sum_{\eta \in \Delta_{k-1}(K_h)} \left(\frac{|\star \eta|}{|\eta|} \right)^{-1} \langle \star_h \omega_h, \partial \star \eta \rangle^2 \\ &= \sum_{\eta \in \Delta_{k-1}(K_h)} \sum_{\substack{\tau \in \Delta_k(K_h), \\ \tau \succ \eta}} \left(\frac{|\star \eta|}{|\eta|} \right)^{-1} \langle \star_h \omega_h, \star \tau \rangle^2 \\ &\leq Ch^{-2} \|\star_h \omega_h\|_h. \end{aligned} \quad (12)$$

Finally, by combining the estimates (9), (10), (11) and (12), with Corollaries 3.3, 3.4, 3.5 and 3.6, we infer

$$\begin{aligned} \|\star_h d_h (\star_h R_h - R_h \star) d\omega\|_h &\leq \|\star_h\|_{\mathcal{L}((C^k(*K_h), \|\cdot\|_h), (C^{n-k}(K_h), \|\cdot\|_h))} \\ &\quad \times \|d_h (\star_h R_h - R_h \star) d\omega\|_h \\ &\lesssim h^{-1} \|(\star_h R_h - R_h \star) d\omega\|_h \\ &\lesssim h^{-1} h = O(1), \\ \|\star_h d_h (\star_h R_h - R_h \star) d\omega\|_\infty &\lesssim h^{2k-n} \|d_h (\star_h R_h - R_h \star) d\omega\|_\infty \\ &\lesssim h^{2k-n} \|(\star_h R_h - R_h \star) d\omega\|_\infty \\ &\lesssim h^{2k-n} h^{n-k} = O(h^k), \end{aligned}$$

$$\|(\star_h R_h - R_h \star) d\star d\omega\|_h = O(h),$$

and

$$\|(\star_h R_h - R_h \star) d\star d\omega\|_\infty = O(h^{k+1}).$$

We have proved the following.

Corollary 3.7 *If K_h is regular, then*

$$\begin{aligned} \Delta_h R_h \omega - R_h \Delta \omega &= \star_h d_h (\star_h R_h - R_h \star) d\omega + (\star_h R_h - R_h \star) d\star d\omega \\ &= O(1) + O(h), \end{aligned}$$

for $\omega \in C^3 \Lambda^0(\mathcal{P})$, in both the maximum and discrete L^2 -norm.

Note that in the displayed equation above, we have used the notation $O(1) + O(h)$ to convey information on the individual sizes of the two terms appearing in the right-hand side of the first line. In fact, the preceding corollary will not be employed in what follows, since it yields the consistency error of $O(1)$. However, the techniques we have used to estimate the individual terms and operator norms will prove to be fruitful.

4 Stability and Convergence

We carry on with a proof of the first order convergence estimates

$$\|u_h - R_h u\|_h = O(h) \quad \text{and} \quad \|d(u_h - R_h u)\|_h = O(h),$$

previously stated in (3). The demonstration is based on a variational formulation—in the *discrete* inner product $(\cdot, \cdot)_h$ —of the Poisson problem (1), whose expression can be derived efficiently using Proposition 2.6.

We put forth a discrete Poincaré-like inequality for 0-cochains that will prove to be essential to the argument. The inequality is established by comparing the discrete norm of these cochains with the continuous L^2 -norm of Whitney forms

$$\phi_\tau = k! \sum_{i=0}^k (-1)^i \lambda_i d\lambda_1 \wedge \cdots \wedge \widehat{d\lambda_i} \wedge \cdots \wedge d\lambda_k, \quad \tau \in \Delta_k(K_h),$$

where λ_i is the piecewise linear hat function evaluating to 1 at the i th vertex of τ and 0 at every other vertices of K_h . Setting

$$W_h \omega_h = \sum_{\tau} \omega_h(\tau) \phi_\tau \quad \text{for all } \tau \in \Delta_k(K_h),$$

for all $\omega_h \in C^k(K_h)$ defines linear maps W_h , called Whitney maps, which have the key property that $W_h d_h \omega_h = d W_h \omega_h$.

Theorem 4.1 *Let K_h be a family of regular triangulations. There exist two positive constants c_1 and c_2 , independent of h , satisfying*

$$c_1 \|\omega_h\|_h \leq \|W_h \omega_h\|_{L^2 \Lambda^k(K_h)} \leq c_2 \|\omega_h\|_h, \quad \omega_h \in C^k(K_h).$$

Proof We proceed using a scaling argument. Suppose $\sigma = [v_0, \dots, v_n] \in \Delta_n(K_h)$, and assume $\tau < \sigma$ is an orientation of the face $v_0, \dots, v_k$. Let $\{\lambda_i\}$ be the barycentric coordinate functions associated to the vertices of σ (these are the piecewise linear hat functions used in the definition of Whitney forms). The 1-forms $d\lambda_0, \dots, d\lambda_\ell, \dots, d\lambda_n$ are a basis for $\text{Alt}^n(\mathbb{R}^n)$ dual to $t_i^\ell = v_i - v_\ell, i = 1, \dots, n$, i.e. the vectors $t_1^\ell, \dots, t_n^\ell$ form a basis for $P(\sigma)$ and satisfy $d\lambda_1 \wedge \dots \wedge \widehat{d\lambda_\ell} \wedge \dots \wedge d\lambda_n(t_1^\ell, \dots, t_n^\ell) = 1$. In this setting, it follows in general that for any oriented face $\rho = [v_{\rho(0)}, \dots, v_{\rho(m)}]$ of σ ,

$$\begin{aligned} |\rho| &= |\det(v_{\rho(1)} - v_{\rho(0)}, \dots, v_{\rho(m)} - v_{\rho(0)})| \\ &= |\det(t_1^{\rho(0)}, \dots, t_m^{\rho(0)})| = \pm_\rho \frac{1}{m!} \text{vol}_\rho(t_1^{\rho(0)}, \dots, t_n^{\rho(0)}). \end{aligned}$$

In other words,

$$d\lambda_{\rho(1)} \wedge \dots \wedge \widehat{d\lambda_{\rho(0)}} \wedge \dots \wedge d\lambda_{\rho(m)} = \pm_\rho \frac{1}{|\rho|m!} \text{vol}_\rho, \quad (13)$$

where $\pm_\rho$ depends on the orientation of ρ .

Now let $\hat{\sigma}$ denote the standard n -simplex in $\mathbb{R}^n$, and consider an affine transformation $F: \hat{\sigma} \rightarrow \sigma$ of the form $F = B + b$, where $B: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a linear map and $b \in \mathbb{R}^n$ a vector. We compute the pullback $\hat{\phi}_\tau = F^* \phi_\tau$ of a Whitney k -form ϕ_τ using (13). To lighten notation, we write $\hat{\tau} = F^{-1}(\tau)$ and $B_{\hat{\tau}} = B|_{P(\hat{\tau})}$. We evaluate directly

$$\begin{aligned} (F^* \phi_\tau)_x &= k! \sum_{i=0}^k \frac{\pm_i \lambda_i \circ F}{k!|\tau|} F^* \text{vol}_\tau \\ &= k! \sum_{i=0}^k (-1)^i (\lambda_i \circ F) |\det B_{\hat{\tau}}| \bigwedge_{\substack{\ell=0 \\ \ell \neq i}}^k d\lambda_\ell \\ &= |\det B_{\hat{\tau}}| (\phi_\tau)_{F(x)}. \end{aligned}$$

A change of variables then yields

$$\begin{aligned} \int_\sigma \left\| \sum_\tau \omega_h(\tau) \phi_\tau \right\|^2 \text{vol}_\sigma &= \int_\sigma \left\| \sum_\tau \frac{\omega_h(\tau)}{|\det B_{\hat{\tau}}|} (\hat{\phi}_\tau)_{F^{-1}} \right\|^2 \text{vol}_\sigma \\ &= |\det B| \int_{\hat{\sigma}} \left\| \sum_\tau \frac{\omega_h(\tau)}{|\det B_{\hat{\tau}}|} \hat{\phi}_\tau \right\|^2 \text{vol}_{\hat{\sigma}}. \end{aligned}$$

Using the equivalence of norms on finite dimensional Banach spaces and the regularity assumption on K_h , we therefore conclude that

$$\|W_h \omega_h\|_{L^2 \Lambda^k(\sigma)}^2 \sim |\det B| \sum_{\tau} \left(\frac{\omega_h(\tau)}{|\det B_{\hat{\tau}}|} \right)^2 \sim h^{n-2k} \sum_{\tau} \omega_h(\tau)^2 \sim \|\omega_h\|_h^2,$$

where the equivalences do not depend on h . $\square$

Corollary 4.2 *There exists a constant C , independent of h , such that the discrete Poincaré inequality*

$$\|\omega_h\|_h \leq C \|d_h \omega_h\|_h$$

holds for all $\omega_h \in C^0(K_h)$ such that $\omega_h = 0$ on ∂K_h .

Proof Using Theorem 4.1 and the Poincaré inequality, we have

$$\begin{aligned} \|\omega_h\|_h &\lesssim \|W_h \omega_h\|_h \lesssim \|d W_h \omega_h\|_{L^2 \Lambda^k(K_h)} \\ &= \|W_h d_h \omega_h\|_{L^2 \Lambda^k(K_h)} \lesssim \|d_h \omega_h\|_h, \end{aligned} \quad (14)$$

which establishes the proof. $\square$

The argument is twofold. We first restrict our attention in (1) to the homogeneous boundary condition $g = 0$, and introduce in a second breath inhomogeneous Dirichlet conditions. Consider the following variational formulations.

Suppose that $v_h \in C^0(K_h)$ vanishes everywhere but at a vertex p . For all $\omega_h \in C^0(K_h)$, we have

$$(\delta_h d_h \omega_h, v_h)_h - (R_h f, v_h)_h = v_h(p) |*p| (\langle \delta_h d_h \omega_h, p \rangle - \langle R_h f, p \rangle).$$

In other words, $\Delta_h \omega_h = R_h f$ if and only if $(\delta_h d_h \omega_h, v_h)_h = (R_h f, v_h)_h$ for all v_h . By Proposition 2.6, we may thus equivalently find a discrete function ω_h vanishing on ∂K_h such that

$$(d\omega_h, d_h v_h) = (R_h f, v_h)_h \quad (15)$$

for all 0-cochains v_h satisfying the homogeneous boundary condition. Note that this problem in turn reduces to the one of minimizing the energy functional

$$E_h(v_h) = \frac{1}{2} (d_h v_h, d_h v_h)_h - (R_h f, v_h)_h$$

over the same collection of v_h . Indeed a direct computation shows that ω_h satisfies (15) if and only if $\frac{d}{d\epsilon}\big|_{\epsilon=0} E_h(\omega_h + \epsilon v_h) = 0$ for all these v_h , in which case we further conclude from

$$E_h(\omega_h + v_h) = \underbrace{\frac{1}{2} (d_h \omega_h, d_h \omega_h)_h}_{\geq 0} + \underbrace{(\omega_h, v_h)_h - (R_h f, \omega_h)_h}_{=0 \text{ by (15)}} + \underbrace{\frac{1}{2} (v_h, v_h)_h - (R_h f, v_h)_h}_{E_h(v_h)}$$

that ω_h is a minimizer. Since $(d_h u_h, d_h u_h)_h = 0$ if and only if u_h is constant, $\delta_h d_h$ is invertible over the space of discrete functions vanishing on ∂K_h . The existence and uniqueness of a solution to (15) is thus guaranteed for all h .

Using Corollary 4.2 in (15) and assuming $R_h f$ is not identically 0 (the case with trivial solution $\omega_h = 0$), we obtain

$$(d_h \omega_h, d_h \omega_h)_h \leq \|\omega_h\|_h \|R_h f\|_h \leq C \|d_h \omega_h\|_h \|R_h f\|_h,$$

and another application of that corollary finally yields the stability estimate

$$\|\omega_h\|_h \leq C \|R_h f\|_h.$$

Unfortunately, convergence cannot be obtained with an application of the Lax–Richtmyer theorem with this estimate and the consistency bound derived in Corollary 3.7. However, Proposition 2.6 comes to our rescue. Given a solution ω to the associated continuous problem, it allows us to use the expression for Δ_h given in Lemma 3.1 to evaluate the error $e_h = R_h \omega - \omega_h$ with

$$\begin{aligned} (d_h e_h, d_h e_h)_h &= (\Delta_h e_h, e_h)_h \\ &= (\star_h d_h (\star_h R_h - R_h \star) d\omega, e_h)_h + ((\star_h R_h - R_h \star) d\star d\omega, e_h)_h \\ &= (\star_h^{-1} (\star_h R_h - R_h \star) d\omega, d_h e_h)_h + ((\star_h R_h - R_h \star) d\star d\omega, e_h)_h \\ &\leq Ch \|d_h e_h\|_h + Ch \|e_h\|_h, \end{aligned} \quad (16)$$

where the last inequality was obtained from an application of Corollary 3.5, (11) and the second to last estimate in Corollary 3.7. Using the discrete Poincaré-like inequality again completes the proof of the homogeneous case of the following theorem.

Theorem 4.3 *The unique discrete solution $\omega_h \in C^0(K_h)$ of problem (1) stated for 0-forms over a regular triangulation K_h satisfy*

$$\|e_h\|_h \leq C \|d_h e_h\|_h = O(h).$$

As one would expect, the case of inhomogeneous boundary conditions reduces to the homogeneous one. Given $g_h \neq \vec{0}$, the Poisson problem is to find ω_h in the affine space $\{g + \eta_h \mid \eta_h \in C^0(K_h), \eta_h = 0 \text{ on } \partial K_h\}$ satisfying (15). That is, we are looking for $u_h \in C^0(K_h) \cap \{\eta_h : \eta_h|_{\partial K_h} = 0\}$ satisfying

$$(\mathbf{d}_h u_h, \mathbf{d}_h v_h)_h = (R_h f, v_h)_h - (\mathbf{d}_h g_h, \mathbf{d}_h v_h)_h = F(u_h, v_h), \quad (17)$$

for all v_h vanishing on the boundary, which is equivalent to the previous homogeneous problem, but with the linear functional $F : C^0(K_h) \times C^0(K_h) \rightarrow \mathbb{R}$ on the right-hand side. Repeating the previous arguments, we obtain the stability estimate

$$\|u_h\|_h \leq C (\|R_h f\|_h + \|\mathbf{d}_h g_h\|_h). \quad (18)$$

The rest of the proof goes through similarly, so that the solution $\omega_h = g_h + u_h$ to (1) with inhomogeneous boundary condition g is first order convergent.

Remark 4.4 The argument (16) can be interpreted as measuring the consistency error in a discrete H^{-1} -type norm (as the dual to an H^1 -type norm), instead of an L^2 -type norm. We could have made this structure more explicit by introducing a discrete H^{-1} -norm, but we prefer the current presentation as it is more straightforward.

5 Numerical Experiments

In this section, we report on some numerical experiments performed over two and three dimensional triangulations. The discrete solutions are computed from the inverse of Δ_h built as a compound of the operators defined in Sect. 2.3. The volumes needed to implement the Hodge stars as diagonal matrices $\star_k^h$ are computed in a standard way. The matrices $\mathbf{d}_{k-1, \text{Primal}}^h = (\partial_k^h)^T$ acting on vector representations of cochains (denoted by $[\cdot]$) in $C^{k-1}(K_h)$ are created using the algorithm suggested in [2]. By construction, the orientation of $\mathbf{d}_{k-1, \text{Primal}}^h[\tau] \in \Delta_k(K_h)$ is therefore consistent with the orientation of the $(k-1)$ -simplex τ . Hence, it is suitably oriented for Definition 2.3 to imply

$$\begin{aligned} (\mathbf{d}_{n-k, \text{Dual}}^h \star_k^h[\omega_h])^T[\tau] &= \langle \mathbf{d}_h \star_h \omega_h, \star \tau \rangle \\ &= (-1)^k \sum_{\eta > \tau} \langle \star_h \omega_h, \star \eta \rangle \\ &= (-1)^k (\star_h^k[\omega_h])^T \mathbf{d}_{k-1, \text{Primal}}^h[\tau] \\ &= (-1)^k ((\mathbf{d}_{k-1, \text{Primal}}^h)^T \star_h^k[\omega_h])^T[\tau] \end{aligned}$$

for all $\omega_h \in C^k(K_h)$ and $\tau \in \Delta_{k-1}(K_h)$. This proves the practical definition

$$\mathbf{d}_{n-k, \text{Dual}}^h = (-1)^k (\mathbf{d}_{k-1, \text{Primal}}^h)^T = (-1)^k \partial_k^h$$

given in [6], which we use in the following.

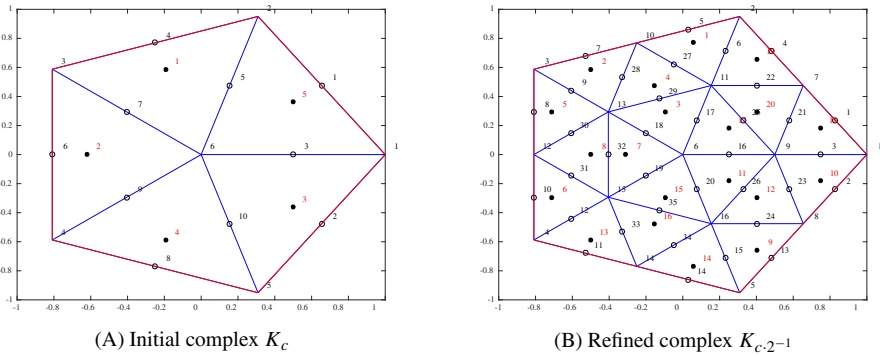


Fig. 4 The initial primal triangulation of W_5 and its first refinement are shown. On the left, (a) was obtained by fixing the length of the interior edges to 1, consequently setting the initial maximal length for the edges to $c = (2 - 2 \cos(2\pi/5))^{1/2}$. The triangles in K_C are subdivided by the edges of their medial triangles to produce (b)

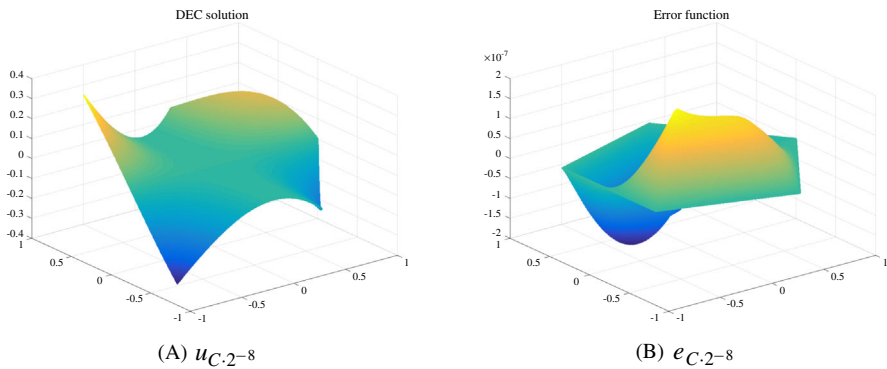


Fig. 5 This figure refers to the experiment $u(x, y) = x^2 \sin(y)$. On the left, a DEC solution to the Hodge–Laplace Dirichlet problem over $T_{C,2^{-8}}$ is displayed. A plot of the related error function is found on the right

5.1 Experiments in Two Dimensions

Convergence of DEC solutions for 0-forms is first studied over two types of convex polygons. Non-convex polygons are then used to evaluate the consequences of a lack of regularity caused by re-entrant corners.

5.1.1 Regular Polygons

We conduct numerical experiments on refinements of the form $\{K_{c,2^{-i}}\}_{i=0}^N$ of the wheel graph W_5 whose boundary is a regular pentagon. Triangulation of the pentagonal domain is performed by recursively subdividing each elementary 2-simplex of the initial complex into its four inscribed subtriangles delimited by the graph of its medial triangle and its boundary. We illustrate the process in Fig. 4.

Table 1 Experiment of Sect. 5.1.1 with $u(x, y) = x^2 \sin(y)$

i	$e_i^\infty = \ e_{C,2^{-i}}\ _\infty$	$\log(e_i^\infty/e_{i-1}^\infty)$	$e_i^{H^1} = \ de_{C,2^{-i}}\ _{C,2^{-i}}$	$\log(H_i^1/e_{i-1}^{H^1})$	$e_i^{L^2} = \ e_{C,2^{-i}}\ _{C,2^{-i}}$	$\log(L_i^2/e_{i-1}^{L^2})$
0	1.561251e-17	–	2.975694e-17	–	1.487847e-17	–
1	3.202794e-03	– 4.754362e+01	1.072846e-02	– 4.835714e+01	2.821094e-03	– 4.743002e+01
2	7.836073e-04	2.031128e+00	2.879579e-03	1.897512e+00	6.332754e-04	2.155350e+00
3	1.956510e-04	2.001848e+00	7.353114e-04	1.969431e+00	1.532456e-04	2.046987e+00
4	4.891893e-05	1.999818e+00	1.849975e-04	1.990850e+00	3.798925e-05	2.012183e+00
5	1.227086e-05	1.995157e+00	4.633277e-05	1.997401e+00	9.477213e-06	2.003057e+00
6	3.067823e-06	1.999949e+00	1.158895e-05	1.999283e+00	2.368052e-06	2.000762e+00
7	7.669629e-07	1.999987e+00	2.897627e-06	1.999806e+00	5.919350e-07	2.000190e+00
8	1.917491e-07	1.999937e+00	7.244331e-07	1.999948e+00	1.479789e-07	2.000047e+00

The terms $C \cdot 2^{-i} \in \mathbb{R}$ indicates the values of h in $\|\cdot\|_h$

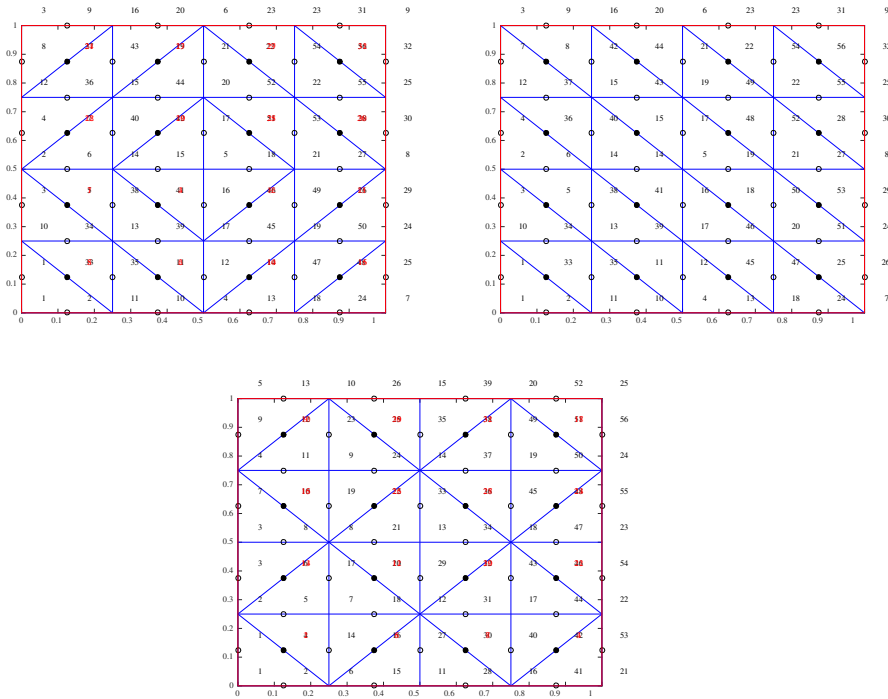


Fig. 6 Three regular triangulations of the square

A discrete solution to the Hodge–Laplace Dirichlet problem with trigonometric exact solution $u(x, y) = x^2 \sin(y)$ over this complex is displayed along with its error function in Fig. 5. The size of the errors for its collection of refinements with $N = 9$ is found in different discrete norms in Table 1. The results show second order convergence both in the discrete L^2 and H^1 norms. Other cases where the initial primal triangulation is designed from a wheel graph on $n + 1$ vertices are also considered. In particular, repeating the experiment on regular n -gons with $6 \leq n \leq 8$ yields the same asymptotics.

5.1.2 Rectangular Domains

Three different types of regular triangulations were used to investigate convergence of DEC solutions over the unit square. These appear as degenerate examples of circumcentric complexes in which the circumcenters lie on the hypotenuses of the primal triangles. Examples of these triangulations are shown in Fig. 6.

The computations carried over the unit square produce results similar to the ones previously obtained in Sect. 5.1.1.

5.1.3 Unstructured Meshes

Second order convergence in all norms is also observed for unstructured meshes on convex polygons. The discrete solutions behaved similarly over non-convex polygons

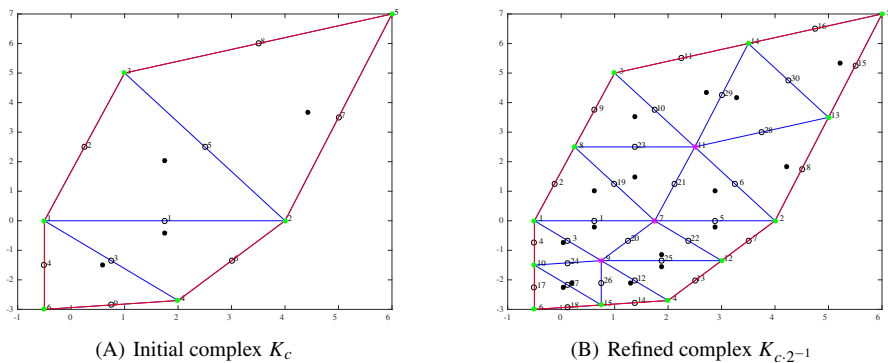


Fig. 7 Unstructured triangulations were refined as in Sect. 5.1.1

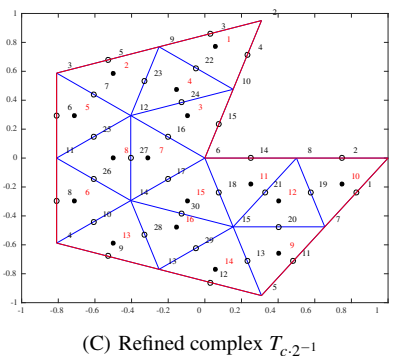
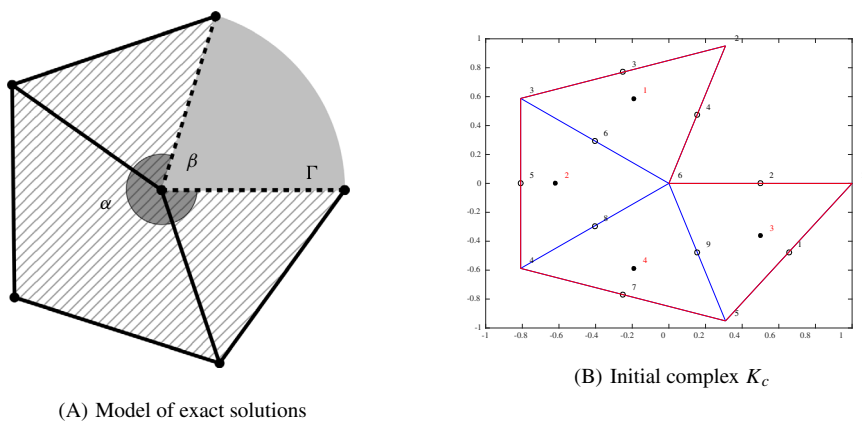


Fig. 8 We design an exact solution $u \in H^{1+\mu}(\mathcal{P})$, $0 < \mu < 1$, such that $\Delta u = 0$ in $\mathcal{P}$ and $u = 0$ on the dotted piece of the boundary Γ drawn in (a). The red boundary shown in (b) and (c) corresponds to a concave cycle of W_5 on which calculations were conducted. This is an instance of the case where $\alpha = 8\pi/5$ in (a). The constant c was fixed as in Fig. 4

Table 2 Experiment of Sect. 5.1.4 with $r^\mu \sin(\mu\theta)$, $\mu = 5/8$

i	$e_i^\infty = \ e_{C,2^{-i}}\ _\infty$	$\log(e_i^\infty/e_{i-1}^\infty)$	$e^{H_d^1} = \ de_{C,2^{-i}}\ _{C,2^{-i}}$	$\log(H_d^1/e_{i-1}^{H_d^1})$	$e^{L_d^2} = \ e_{C,2^{-i}}\ _{C,2^{-i}}$	$\log(L_d^2/e_{i-1}^{L_d^2})$
0	0	–	0	–	0	–
1	3.402738e–02	–	8.467970e–02	–	2.346479e–02	–
2	3.194032e–02	9.131748e–02	6.533106e–02	3.742472e–01	1.353817e–02	7.934654e–01
3	2.346298e–02	4.449927e–01	4.496497e–02	5.389676e–01	6.570546e–03	1.042947e+00
4	1.595752e–02	5.561491e–01	2.983035e–02	5.920204e–01	2.970932e–03	1.145097e+00
5	1.054876e–02	5.971636e–01	1.952228e–02	6.116590e–01	1.299255e–03	1.193231e+00
6	6.894829e–03	6.134867e–01	1.270715e–02	6.194814e–01	5.584503e–04	1.218184e+00
7	4.485666e–03	6.201927e–01	8.252738e–03	6.226958e–01	2.377754e–04	1.231830e+00
8	2.912660e–03	6.229847e–01	5.354822e–03	6.240341e–01	1.007013e–04	1.239517e+00

The terms $C \cdot 2^{-i} \in \mathbb{R}$ indicates the values of h in $\|\cdot\|_h$

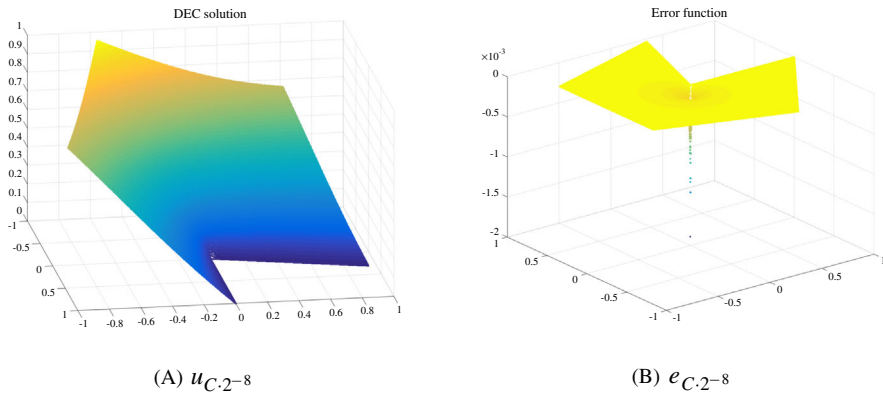


Fig. 9 These plots were obtained from the experiment $r^\mu \sin(\mu\theta)$, $\mu = 5/8$. On the left, a DEC solution to the Hodge–Laplace Dirichlet problem over $K_{C,2-8}$ is displayed. A plot of the related error function is found on the right

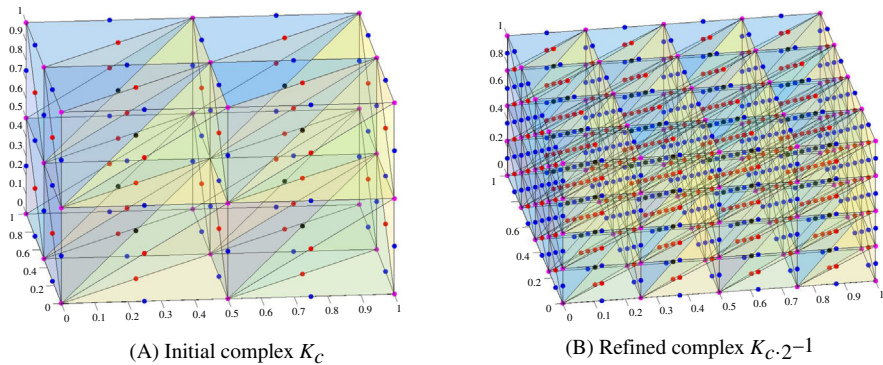


Fig. 10 The initial 3-dimensional primal triangulation and its first refinement are shown. A sequence of finer triangulations was created by gluing at each step 4 smaller copies of the previous refinement into the initial cubic domain

when the solution is smooth enough. One example of a primal domain over which the algorithm was tested is shown in Fig. 7.

5.1.4 Non-convex Polygons

We study the convergence behavior of DEC when a reentrant corner is present along the boundary of the primal complex. More precisely, we consider re-entrant corners with various angles, leading to the exact solutions $u \in H^{1+\mu}(\mathcal{P})$, $0 < \mu < 1$, which lack H^2 regularity.

Elected candidates for $u \in H^{1+\mu}(\mathcal{P})$ were harmonic functions of the form $r^\mu \sin(\mu\theta)$. These functions suit the model of Fig. 8 (A) for $\mu = \pi/(2\pi - \beta) = \pi/\alpha$ and the result of the experiments can be found in Table 2. The initial triangulation for this case is also plotted along with its first refinement in Fig. 8. The refinement algorithm

Table 3 Experiment of Sect. 5.2 with $u(x, y) = x^2 \sin(y) + \cos(z)$

i	$e_i^\infty = \ e_{C,2^{-i}}\ _\infty$	$\log(e_i^\infty/e_{i-1}^\infty)$	$e_d^{H^1} = \ de_{C,2^{-i}}\ _{C,2^{-i}}$	$\log(H_d^1/e_{i-1}^{H^1})$	$e_d^{L^2} = \ e_{C,2^{-i}}\ _{C,2^{-i}}$	$\log(L_d^2/e_{i-1}^{L^2})$
0	8.586493e-04	–	1.487224e-03	–	3.035784e-04	–
1	2.666725e-04	1.687000e+00	6.216886e-04	1.258358e+00	1.156983e-04	1.391702e+00
2	7.122948e-05	1.904523e+00	1.774812e-04	1.808526e+00	3.166206e-05	1.869540e+00
3	1.835021e-05	1.956678e+00	4.594339e-05	1.949737e+00	8.083333e-06	1.969733e+00
4	4.621759e-06	1.989283e+00	1.158904e-05	1.987096e+00	2.031176e-06	1.992635e+00

The terms $C \cdot 2^{-i} \in \mathbb{R}$ indicates the values of h in $\|\cdot\|_h$

of Sect. 5.1.1 was used. Other experiments were performed and led to the empirical observation that $\|u - u_h\|_h \sim h^{2\mu}$ and $\|d(u - u_h)\|_h \sim h^\mu$. This convergence behavior is precisely what would be expected when a finite element method is applied to the same problem, which may be explained by the fact that our discretization is equivalent to a finite element method, cf. [8,22] (Fig. 9).

5.2 Three-Dimensional Case

As explained in [18], it is not an easy task in general to generate and refine tetrahedral meshes of 3-dimensional arbitrary convex domains without altering its circumcentric quality. The next example is a degenerate case: the circumcenters of the tetrahedra used to triangulate the cube lies on lower dimensional simplices, but it does allow for a better control over the step size in h (and thus for a better evaluation of the convergence rate), because the mesh can be refined recursively by gluing smaller copies of itself filling the cuboidal domain. The triangulation is displayed in Fig. 10. Again, a second order convergence in all norms similar to the results of Sect. 5.1.1 is obtained and presented in Table 3.

6 Conclusion

This paper fills a gap in the current literature, which has yet to offer any theoretical validation regarding the convergence of discrete exterior calculus approximations for PDE problems in general dimensions. Namely, we prove that discrete exterior calculus approximations to the scalar Poisson problem converge at least linearly with respect to the mesh size for quasi-uniform triangulations in arbitrary dimensions. Nevertheless, it must be emphasized that this *first order* convergence result is only partially satisfactory. In accordance with [14], the numerical experiments of Sect. 5 display pointwise *second order* convergence of the discrete solutions over unstructured triangulations. The same behavior was observed also for the discrete L^2 -norm. The challenges of explaining the second order convergence, and of proving convergence for general p -forms persist.

Due emphasis must be given to the role played by compatibility in obtaining this result. The reliable framework of the continuous setting was reproduced at the discrete level to a sufficient extent so as to successfully provide a theorem of stability comparable to the one found in the classical PDEs literature. This makes another convincing case for the study and development of structure preserving discretization in general.

An accessory consequence of our investigations is the technical modification of the combinatorial definition for the boundary of a dual cell currently found in the early literature. We suggest that it is presently only compatible with the continuous theory up to a sign, and that accounting for the latter yields a new definition which agrees with the algorithms later described in [6]. A compatible extension of the usual discrete L^2 -norm to $C^k(*K_h)$ was also explicitly introduced.

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References

1. Arnold, D.N., Falk, R.S., Winther, R.: Finite element exterior calculus, homological techniques, and applications. *Acta Numerica* **15**, 1–155 (2006)
2. Bell, N., Hirani, A.N.: PyDEC: software and algorithms for discretization of exterior calculus. *ACM Trans. Math. Softw.* **39**(1), 41 (2012)
3. Ciarlet, P.G.: *Finite Element Method for Elliptic Problems*. Society for Industrial and Applied Mathematics, Philadelphia, PA (2002)
4. Crane, K., de Goes, F., Desbrun, M., Schroder, P.: Digital Geometry Processing with Discrete Exterior Calculus. In: *Proceedings of ACM SIGGRAPH 2013 Courses*. SIGGRAPH '13. ACM, New York, NY, USA (2013)
5. Desbrun, M., Hirani, A.N., Leok, M., Marsden, J.E.: Discrete Exterior Calculus. [arXiv:math/0508341](https://arxiv.org/abs/math/0508341) (2005)
6. Desbrun, M., Kanso, E., Tong, Y.: Discrete Differential Forms for Computational Modeling. In: Bobenko, A.I., Sullivan, J.M., Schröder, P., Ziegler, G.M. (eds.) *Discrete Differential Geometry*, pp. 287–324. Birkhäuser, Basel (2008)
7. Frauendiener, J.: Discrete differential forms in general relativity. *Class. Quantum Gravity* **23**(16), S369 (2006)
8. Hildebrandt, K., Polthier, K., Wardetzky, M.: On the convergence of metric and geometric properties of polyhedral surfaces. *Geom. Dedic.* **123**, 89–112 (2006)
9. Hiptmair, R.: Discrete Hodge operators. *Num. Math.* **90**(2), 265–289 (2001)
10. Hirani, A.N.: Discrete exterior calculus. Thesis (Ph.D.), California Institute of Technology. ProQuest LLC, Ann Arbor, MI, p. 103 (2003)
11. Hirani, A.N., Kalyanaraman, K., VanderZee, E.B.: Delaunay Hodge star. *Comput. Aided Des.* **45**(2), 540–544 (2013)
12. Hirani, A.N., Nakshatrala, K.B., Chaudhry, J.H.: Numerical method for darcy flow derived using discrete exterior calculus. *Int. J. Comput. Methods Eng. Sci. Mech.* **16**(3), 151–169 (2015)
13. Mohamed, M.S., Hirani, A.N., Samtaney, R.: Discrete exterior calculus discretization of incompressible Navier–Stokes equations over surface simplicial meshes. *J. Comput. Phys.* **312**, 175–191 (2016)
14. Nong, H.D.: Numerical Study of Discrete Laplace–Beltrami. Tech. Rep. California Institute of Technology (2004)
15. Seslija, M., van der Schaft, A., Scherpen, J.M.A.: Discrete exterior geometry approach to structure-preserving discretization of distributed-parameter port-Hamiltonian systems. *J. Geom. Phys.* **62**(6), 1509–1531 (2012)
16. Stern, A., Tong, Y., Desbrun, M., Marsden, J.E.: Geometric computational electrodynamics with variational integrators and discrete differential forms. *Fields Inst. Commun.* **73**, 437–475 (2007)
17. Stern, A., Tong, Y., Desbrun, M., Marsden, J.E.: Variational integrators for Maxwell’s equations with sources. [arXiv:0803.2070](https://arxiv.org/abs/0803.2070) (2008)
18. VanderZee, E., Hirani, A.N., Guoy, D.: Triangulation of simple 3D shapes with well-centered tetrahedra. In: Rao, V.G. (ed.) *Proceedings of the 17th International Meshing Roundtable*, pp. 19–35. Springer, Berlin (2008)
19. VanderZee, E., Hirani, A.N., Guoy, D., Ramos, E.: A.: Well-centered triangulation. *SIAM J. Sci. Comput.* **31**(6), 4497–4523 (2009/2010)
20. VanderZee, E., Hirani, A.N., Zharnitsky, V., Guoy, D.: A dihedral acute triangulation of the cube. *Comput. Geom.* **43**(5), 445–452 (2010)
21. VanderZee, E., Hirani, A.N., Guoy, D., Zharnitsky, V., Ramos, E.A.: Geometric and combinatorial properties of well-centered triangulations in three and higher dimensions. *Comput. Geom.* **46**(6), 700–724 (2013)
22. Wardetzky, M.: Convergence of the Cotangent Formula: An Overview. *Discrete Differential Geometry*, pp. 275–286. Oberwolfach Semin, Birkhäuser, Basel (2008)

23. Whitney, H.: Geometric Integration Theory. Princeton University Press, Princeton (1957)
24. Xu, G.: Convergence of discrete Laplace–Beltrami operators over surfaces. *Comput. Math. Appl.* **48**(3–4), 347–360 (2004)
25. Xu, G.: Discrete Laplace–Beltrami operators and their convergence. *Comput. Aided Geom. Des.* **21**(8), 767–784 (2004)
26. Yavari, A.: On geometric discretization of elasticity. *J. Math. Phys.* **49**(2), 022901 (2008)

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